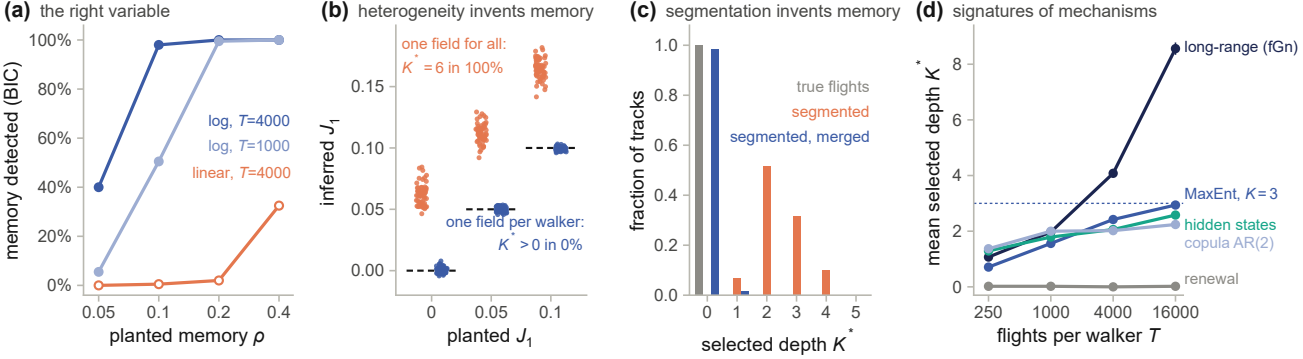

(a) the right variable
memory detected (BIC)
100%
80%
60%
40%
20%
0%
log, T=4000
log, T=1000
linear, T=4000
0.05
0.1
0.2
0.4
planted memory ρ
(b) heterogeneity invents memory
inferred J1
0.15
0.10
0.05
0.00
one field for all:
K* = 6 in 100%
one field per walker:
K* > 0 in 0%
0
0.05
0.1
planted J1
(c) segmentation invents memory
fraction of tracks
1.0
0.8
0.6
0.4
0.2
0.0
true flights
segmented
segmented, merged
0
1
2
3
4
5
selected depth K*
(d) signatures of mechanisms
mean selected depth K*
8
6
4
2
0
long-range (fGn)
MaxEnt, K = 3
hidden states
copula AR(2)
renewal
250
1000
4000
16000
flights per walker T